An insight into power system resilience and its confounding traits

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***Abstract*—The frequency of extreme weather events has dramatically triggered numerous power outages in recent years. Therefore, the concept of power system resilience and its enhancement efforts gaining significant popularity among power engineers. This paper aims at investigating the resilience of power systems. Based on the existing literature, we provide a comprehensive review of current practices for the resilience of power systems from various perspectives. Firstly, several definitions and confounding characteristics of resilience in the power system domain are described. Furthermore, various curves for resilience and quantitative metrics were discussed followed by a summary of evaluation and enhancement strategies for power systems. The study provides a way of analysis to aid in the formulation of definitions, metrics, evaluation techniques, and improvement initiatives that are accepted worldwide.**

***Keywords*—power system resilience; resilience metrics; high impact low-frequency events; extreme events**

I. INTRODUCTION

Destructive power system catastrophes are referred to those caused by excessive outages. Power outages can be triggered by severe weather events, leading to significant losses [1]. The occurrence of dangerous events (like storms, seismic attacks, floods etc.) and attacks by humans (like cyber-physical attacks) has risen intensely in the past few years. Power systems have been badly affected by these events in form of long outage times to destruction of key equipment (e.g., transmission cables, power plants, substations etc.) [2]. Hence, efforts for assessment and improvement of the resilience of power systems by power grid operators and researchers are gaining significant interest. The power system resilience PSR concept focuses on low-probability and high-impact events, for instance, a storm or a tornado, heavy snow, floods etc. This demands expansion of controlling, planning and operational strategies to enhance grid resilience against mentioned events. The foremost move towards this objective is development of evaluation methods and resilience metrics (RMs) to assess operation and planning alternatives and to offer technical and economic explanations for enhancing resilience [2].

The word ‘‘resilience’’ in power system has various qualities varying from the capability to ‘‘resist” and ‘‘recover” from a disruptive event to the capability to react to possible disruptive events and freshly developing risks [3], [4], [5]. Although various metrics have been proposed till now for determining the resilience of a power system, still no such definition is standardized yet [3]. Hence, there is a critical need for a detailed review of present procedures and challenges, and for thorough, concrete, and productive suggestions and proposals to add for developing commonly accepted definitions, metrics, and evaluation methods.

It has been bothering for power and energy engineers to converge on a universal definition of PSR. Numerous research teams from various organizations have come collectively to create a universally accepted definition. Resilience is defined by the United Kingdom Energy Research Center (UKERC) [6] as ‘‘the capacity of an energy system to tolerate disturbance and to continue to deliver affordable energy service to consumers’’. Resilience has also been stated as to anticipate, absorb, and quickly recover from low-frequency high impact events [7]. The resilience of the power system is defined in terms of various aspects like resourcefulness, robustness, adaptability, and quick recovery [8]. The same properties also form the basis for the adoption of resilience definition by North America Electric Reliability Corporation (NERC) in [9]. PSR can be understood as the system’s capability of reducing the chances of wide power outages caused by HILP events [10].

One of the traits of PSR is robustness which means the power system can take up a shock and remain in function. The ability to efficiently manage the crisis as it onsets is another feature of PSR and is termed as resourcefulness. After efficient management, rapid or quick recovery for quick restoration of service to normal state is to be followed. Finally, the capability to gain new lessons from past events utilizing new technology and tools comes under the adaptability trait of PSR. According to the UK’s Cabinet Office, for infrastructure resilience the four basic components are discussed [11]. Resistance of infrastructure resilience can be understood as the capability to withstand the events and its effect. Reliability of infrastructure means ensuring design such that it functions under a specified range of conditions and able to reduce the damage. Redundancy refers to the elements which deals with design, planning and system’s capacity to ensure service continuity during a disruption event. Response and recovery components allow quick and efficient response and healing

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from a disrupting incident. which is done by in advance exercises.

In the event of High-Impact, Low-Frequency (HILF) emergencies, the resilience of the power grid varies with the system's ability to maintain an uninterrupted supply to critical loads. When the system encounters any disturbance, it must get back to its initial or normal state within a short period of time, to be resilient [12]. There are certain factors that perform a straight role in deciding the power grid's resilience against calamities [2]. Some of these factors along with their definitions are provided in Table 1.

TABLE I. FACTORS FOR GRID RESILIENCE

Factors	Definition
Number of Overlapped Branches	Sum of common branches in each Combination of path Without Loop in a Feasible Network
Redundancy in Paths	Sum of available paths to the Sum of loads in each Feasible Network
Sources are repeated	Ratio of total existing sources responsible for delivering all loads in each Feasible Network to the total amount of loads.
Operations related to switching	Changes in the status of all switches, making distinct Feasible Networks with no loop and linking each load to sources.
Availability Probability and Factor of Penalty	Based on reliability or probability of source availability, and the distribution of Penalty Factor losses.
Dominance at Aggregated Central Point	Describes the information of the node's importance with the required network topography.

The five prime determinants of grid resilience are event-based, performance-based, implementation based, temporality based and interdependent systems. Cyber threats, Human-caused environmental and physical hazards belong to the Event-based category. The resilience from direct or indirect risks to the system falls under the Performance-based category. Implementation based category covers the planning and operational phase which includes hardening, control strategies etc. The most highly used classification, Temporality based grid resilience includes ability to reduce threat before a disaster, capacity of a system to take up and withstand or survive a hazard, and system's healing, renewal, reconfiguration, and repair after the threat. Finally, the Interdependent systems category covers resilience linked to the economy, human resources, supply chain etc.

II. POWER SYSTEM RESILIENCE METRICS

A. Curves For Power System Resilience:

As of now, there are very few studies focusing precisely on the resilience curve. There are several methods by which researchers can explain the resilience state which is drawn from a variety of sources [13].

a) Resilience triangle

The importance of the resilience triangle as shown in Fig. 1, is in calculating variation between anticipated and actual system performance and transforming this divergence in various ways [14]. A resilience triangle model suggested by MCEER (Multidisciplinary Center for Earthquake Engineering Research) in [15] was used extensively to measure earthquake resilience. Though, this resilience curve presumes a steep and speedy degradation of network performance level which makes it particular to threat. Such a steep drop in network performance exclusively arises during seismic disasters, that last up to seconds or minutes. But the situation is different for weather-related havocs where the duration may last from hours to days [11]. A shortcoming of the resilience triangle is that it rules out the degraded state which follows instantly after the degradation state. For example, when a power shutdown arises because of an event, it is almost impossible to instantly perform system restoration and return it back to normal post event. It needs some time to begin the recovery course. This drawback is resolved with resilience trapezoid, as resilience trapezoid include a post-event degraded state after the catastrophe occurs [12].

b) Multistate curves

An event's resilience curve was proposed by resilience level versus time plot in order to overcome the shortcomings of resilience triangle as shown in Fig. 2 [7]. The curve is split into 5 distinct states to indicate '4Rs' (rapidity, redundancy, resourcefulness, and robustness) of resilience suggested by MCEER [16]. The curve tells the significance of operational measures as being time efficient over infrastructure hardening measures which takes comparatively longer time [11].

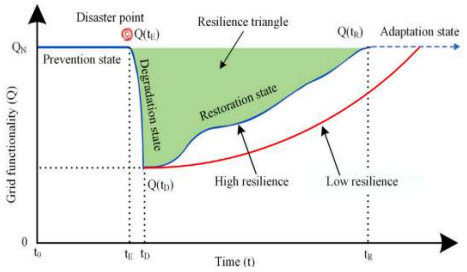


Fig. 1. Resilience triangle with 4 stages during a natural disaster [12]

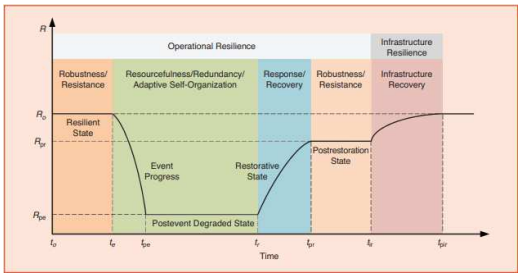


Fig. 2. A theoretical resilience curve related to an event [7].

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c) Resilience trapezoid

Resilience trapezoids as shown in Fig. 3 explain the state of resilience during disrupting events with the help of three phases as discussed below [13] –

During Phase-I: On the occurrence of an event, the network state goes into disruption.

During Phase-II: To meet the load demand emergency coordination is necessary.

During Phase III: Restorative actions take place to reach the normal operating state.

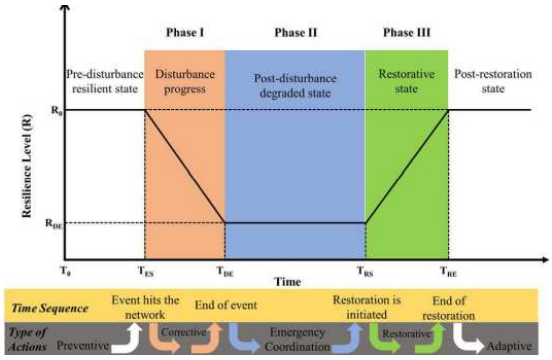


Fig. 3. Resilience trapezoid curve related to an event [11]

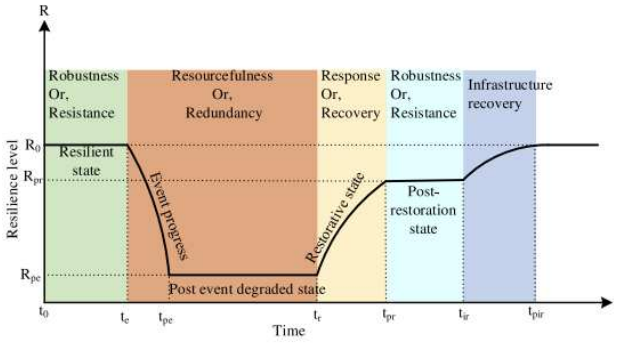


Fig. 4. Resilience curve [12]

B. Power System Resilience Metrics

The RMs for power metrics serve as a tool to gauge the degree of resilience in a power system [17]**.** Although various metrics have been suggested till now for determining PSR, but still no such definition is standardized yet [3]. It is even now a great topic of debate as to how to standardize a RMs [2]. The RMs can be classified in terms of their general characteristics as follows [17], [3]:

a) Metrics based on Attributes give solutions to make up the system further resilient (or resilient to a lesser extent) than system's present status [2]. For instance, numerous system characteristics such as the ability to adapt, recover, resourcefulness, and robustness are quantified using this metric.

b) Metrics based on Performance tell us how resilient a particular power system is. These metrics explain quantifiable information which describes outputs of infrastructure, specification of disturbances, and metrics formulation for infrastructure resilience [2]. RMs based on Performance include the following metrics [5]:

i. *Resilience triangle:* Metrics based on the resilience triangle are simple and easy, but they fail to tackle post-threat degraded state.

ii. *Availability based:* These metrics account for redundancy and rate of reparation but the drawback is that it has a great error margin [18].

iii. *Social welfare-based:* Najafi introduced a resilience metric (RM) that considers the exposure of a population to power and energy, constructed on the social welfare index of hurricanes [19]. But this metric failed to take multi-level complexity of power system.

iv. *Resilience trapezoid:* Metrics based on resilience trapezoid separate infrastructure resilience and operational resilience but these were unsuccessful to indicate post restoration state [13].

v. *Resilience Curve:* The metrics based on resilience curve shown in Fig. 5 [12] provide the primary traits that need to be established by a grid to tackle the varying conditions related to a disaster. These metrics consider the post restoration state, but the computational complexity is high.

vi. *FLEP metrics / Curve-based metrics:* A group of metrics, abbreviated as FLEP metrics, have been presented in [12], [20],[21] depending on specified phases and states of resilience trapezoid. These metrics stand for: how much fast (F) and how much low (L) resilience declines in disturbance progress phase (phase I); how much extensive (E) the post-disturbance degraded state is in phase II (post-disturbance degradation); and how promptly (P) the network recovers in phase III (restorative) [2]. These metrics are numerical based but fail to indicate post restoration state.

- *Φ-metric* (Threat progress State – Phase I) measures the Rate of degradation by determining the gradient of resilience deterioration during the event.
- *Λ-metric* (Threat progress State – Phase I) measures the Level or degree of degradation.
- *E-metric* (Post threat degraded State – Phase II) measures the degradation Extensiveness. It determines the time of this state or phase.
- *Π-metric* (Restorative State – Phase III) measures the rate of recovery by determining the gradient of infrastructure and operational recovery curves. It reflects both the level of initial pre-event resilience and the time taken to enter this level.

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vii Specific Metrics: This is another category of performance-based metrics and is dependent on specific quantities. These are classified as follows[17] :

- *Power Metrics:* Some of the notable power metrics are linked to supplied / unsupplied load [21], mismatch in load [22], available generation capacity [21], etc.
- *Duration Metrics:* Some of the remarkable duration metrics are related to Load or energy curtailment duration [23],[24], STorm Average Interruption Duration Index (STAIDI) [23], Loss Of Load Expectation (LOLE) [26] etc.
- *Frequency Metrics:* These metrics are linked to the frequency of customer disconnections [23], Loss Of Load Frequency (LOLF) [26], total disconnected customers [23], the number of connected customers[27] etc.
- *Probability Metrics:* Loss of Load Probability (LOLP) [28], probability weather the supplied load amount exceeds a certain quantity (after a disaster [29] or specific period [30]) etc. are included in probability metrics.
- *Curve Metrics:* The area between real and ideal performance curves [20], [21] (area of resilience curves [14],[15],[12]) which can be used during the performance degradation phase etc. comes under the Curve metrics category .

III. POWER SYSTEM RESILIENCE EVALUATION

Developing RMs begins with identifying resilience evaluation criteria [2]. The current literature on resilience evaluation generally divides into two categories [14]:

A. Qualitative power system resilience evaluation:

This resilience evaluation can consider the various aspects (such as energy infrastructure, information system, business structure etc.) and various resilience capabilities (like preparedness, mitigation, response, recovery etc.) simultaneously.

B. Quantitative power system resilience evaluation:

Systems performance is often quantified in these resilience evaluations. The evaluation of resilience measures and the comparison of resilience levels of different systems can be enhanced using quantitative metrics. There are three main categories of quantitative resilience evaluation:

a) *Simulation based methods* are most popular because they allow disaster scenarios to be easily incorporated and they can also calculate the effects of disasters. For instance, power flow analysis is adopted in [31] while complex network modeling is adopted in [32].
b) *Analytical method* exploits the likelihood of a failure in a given scenario. For instance, The ability of a network to function as intended despite external factors such as component failure is defined by Whitson and Ramirez-Marquez [33].
c) *Statistical analysis of historic outage data* is meant for data analysis of systems that have stored previous disaster data, records of historical outages and restorations as is done in [34].

IV. RESILIENCE IMPROVEMENT STRATEGIES

Evaluation of resilience forms the basis for resilience enhancement. Resilience evaluations are the quantification methods that can recognize strengths and weaknesses to present resilience enhancing strategies in the power system. Several enhancement approaches being practiced falls under the two broad categories, planning and operational measures [2][11]. The planning measures involve hardening the components of systems to a higher standard. The resilience efficient planning lowers the influence of disaster by reducing the extent of degradation but demands huge investment. On the other hand, for operational measures, the key element is smart grid technology that enables operators to act immediately in the event of a disruption, thereby reducing recovery times and at the same time, these are cost-effective. These measures are basically optimization-based that make use of available assets against failures and severe events [35]. To optimize the characteristics of the utility grid, system planners must carefully select which enhancement measures to implement and balance the different types of measures. This is because an enhancement strategy useful in a specific event may have a destructive impact on another event. For instance, putting power lines underground may make infrastructure more resilient to windstorms, but could take longer to restore in case of a flood or earthquake since it may be harder to trace the fault. Some of the planning and operational measures are given in Table II.

TABLE II. PLANNING AND OPERATIONAL STRATEGIES FOR RESILIENCE

Planning strategies	**Operational strategies**
Upgradation and strengthening of poles and other structures with stronger material and by installing guy wires	Accurate weather forecasting and severity assessment
Managing vegetation	Reconfiguration of network
Doing undergrounding of required transmission or distribution lines	Island formation of microgrid
Installing backup DGs	Management on the demand side
Addition of automatic switches	Mobilization and coordination of repair crew
Rerouting lines to areas that will be less vulnerable to hazards	Installation of mobile energy resources

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V. CONCLUSION

Recently, there has been a noticeable increase in widespread power outages, which implies the systems are not well prepared for high impact events, which used to only be extremely rare. An examination of PSR is presented in this review. Engineers see resilience studies as a relatively new idea to bolster the current power grid against HILP events. In this paper, we examine several definitions for resilience in the power system domain, as well as its confounding characteristics. This is followed by the investigation of several factors and curves for grid resilience. There is also an analysis of distinct metrics for PSR proposed in recent years. In the study, it was observed that there were several overlapping definitions of system resilience provided by different organizations and scholars. This paper is an attempt to contribute to a set of universally accepted and standardized definitions, metrics, and evaluation methods for PSR.

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